BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Security Scenarios

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is penetration testing?

2. What is TCP's role?

A) Choosing display fonts
B) Reliable delivery in order
C) Drawing website images
D) Assigning MAC addresses

3. What is HTTP used for?

4. A pupil logs in to a website. "HTTPS proves every website is honest." Correct this. What does HTTPS protect?

Today's objectives

Recognise relevant networking/security issues in a scenario. Match a threat to an appropriate control.

Key vocabulary - Read, cover, recall

Confidentiality
Keeping information private from unauthorised readers.

Integrity
Keeping data accurate and protected against unauthorised change.

Availability
Making systems and data accessible to legitimate users.

Phishing
Deceptive message seeking sensitive information.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Start with the symptom	Read the scenario and identify what is being targeted: confidentiality, integrity or availability.
Name the threat	Use exact J277 terminology: phishing, malware, brute-force, interception, DoS or SQL injection.
Match the defence	Choose a measure that counters the threat: education, updates, anti-malware, access rights, encryption, authentication, lockout or firewall.
Explain the mechanism	An exam-quality answer says what a control does and why that interrupts the particular attack. Avoid simply listing security products.
Physical security	Physical security uses locked rooms and controlled entry to reduce theft or tampering. Combine this with digital protection such as passwords.

I Do - One worked example

A pupil receives a fraudulent email asking them to sign into a fake school portal. Explain the threat and TWO ways to reduce the risk. [4 marks]

1. The email pretends to be trusted and asks for login credentials.

2. Phishing is a deceptive message intended to steal account details [1].

3. Training users to spot false links and unusual requests [1] can reduce unsafe clicks [1].

4. Two-factor authentication adds another check [1] so a stolen password alone may not allow access. Four teacher marking points are distributed across the explanation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A school account is receiving repeated incorrect passwords. Explain two security controls that limit the attack. [4 marks]

2. A laptop downloads harmful software. Explain two defences that could reduce damage. [4 marks]

3. A school's online service is overloaded with requests. Identify the threat and explain one suitable defence. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change alertLevel to 1. What message prints?

```
alertLevel = 2
if alertLevel > 1 then
print("Tell IT")
else
print("Continue checks")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. A message impersonates a bank and requests a password. Name the threat. [1 mark]

Q3. An employee uses unprotected public Wi-Fi for confidential information. Explain the risk and a suitable control. [4 marks]

Q4. A school faces phishing, malware and brute-force login attempts. Describe one suitable protection for each and how it helps. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Choosing security measures should depend on the particular threat described. ☐ T ☐ F
2. Read the scenario and identify what is being targeted: confidentiality, integrity or availability. ☐ T ☐ F
3. Use exact J277 terminology: phishing, malware, brute-force, interception, DoS or SQL injection. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Testing with permission, within agreed limits, to find and fix security weaknesses. ☐

Q2 B - Reliable delivery in order ☐

Q3 Transferring web resources between web clients and servers. ☐

Q4 HTTPS encrypts data in transit. It does not prove the website or its owner is honest. ☐

Code First - Model

```
alertLevel = 1
if alertLevel > 1 then
print("Tell IT")
else
print("Continue checks")
endif
```

Output: Continue checks

We Do - Check the model

1. An account lockout stops repeated attempts after a threshold [1], limiting brute-force guesses [1]. Two-factor authentication requires another proof of identity [1], so a guessed password alone is insufficient [1]. ☐

2. Anti-malware detects/quarantines the threat [1], reducing malicious activity [1]. Restricted user access rights prevent ordinary accounts making system-wide changes [1], limiting potential damage [1]. ☐

3. It is a denial-of-service attack [1] using excessive traffic/requests [1]. Firewall/rate limiting can filter or restrict unwanted traffic [1], helping keep service available to legitimate users [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Phishing/social engineering [1]. ☐

Q3 Data could be intercepted during transmission [1] and private details read [1]. Encryption protects content in transit [1] so an interceptor without the key cannot understand it [1]. ☐

Q4 Phishing: awareness training [1] reduces risky link clicks [1]. Malware: anti-malware software [1] detects/quarantines malicious programs [1]. Brute-force: account lockout [1] limits repeated password attempts [1]. Other correctly justified measures acceptable. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: